

Evidence Report

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Title: Network Forensics with
TCPdump and Tshark

Case: 26-T004

Date: 03/17/2026

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-03-18	0.1	Draft
Tytus Gonzalez	2026-03-22	1.0	Final

Executive Summary

On 03/05/2026 Professor Stauffer requested the network forensics and analysis of a Linux binary suspected of being malicious. The analysis was conducted under Case Number 26-T004 on my Greyhatter VM with the hypervisor, VMware Workstation Pro. I, Tytus Gonzalez ("Investigator") have been tasked with capturing, preserving, and analyzing all the network traffic after running the 'network-forensics-example-amd64' binary.

The evidence consisted of one Linux executable file, 'network-forensics-example-amd64', designed to simulate malware. The binary was executed on my VM running Fedora Linux 42 to ensure isolation and prevent unintended system impact. Before the execution, all background applications and services on the VM were closed to minimize extra network traffic. Using the tool *tcpdump*, network traffic was captured in real time on the primary network interface and saved as a packet capture file. The capture process started before the execution of the binary and ended immediately after the execution. This process ensured that only relevant traffic activity was recorded. After the packet capture file was created, it was analyzed using *tshark*. I used different techniques to examine the conversations, protocols, endpoint addresses, and DNS queries. This analysis allowed me to determine that there were outbound network connections, performed DNS resolution, and communicated with external servers using UDP. There were also TCP communications conducted over TLS which show that the data traffic was encrypted. There were no HTTP or SMB traffic observed in the analysis.

The purpose of this investigation was to identify and analyze the network behavior of the suspected malware. The network traffic findings indicate that the binary performs typical malware-like actions, such as external communication. All the captured traffic was preserved and stored in the packet capture file 'malware-capture.pcap' which ensures integrity of the analysis.

Synopsis

This assignment involved capturing and analyzing network traffic generated by a simulated malware binary within a virtual environment. I used tools such as tcpdump and tshark to record the network activity and find any patterns, protocols used, and potential IoC. I answered the following questions in the assignment instructions:

1. Looking at IP conversations, what is the destination IP address of the conversation with the highest number of frames in and out? **143.204.130.3**
2. Looking at TCP conversations, what is the destination port number for the IP address in the first question?
13 packets of 443 → HTTPS
14 packets of 50550 → temporary connection
3. Looking at UDP conversations, what protocol is associated with the port found in all conversations? **53 → Domain Name System (DNS)**

4. Provide a sorted list of source IP addresses, with their respective counts.

35 192.168.233.135

14 143.204.130.3

7 142.250.114.99

5 192.168.233.2

4 169.254.144.117

5. Provide a sorted list of destination IP addresses, with their respective counts.

26 192.168.233.135

13 143.204.130.3

11 192.168.233.2

10 142.250.114.99

4 239.255.255.250

1 224.0.0.251

6. Provide a sorted list of DNS queries, with their respective counts.

6 via.placeholder.com

4 www.google.com

4 vqaja7xm18.execute-api.us-east-1.amazonaws.com

2 fedoraproject.org

7. Looking at hosts, what are the IP addresses for the hostnames in the DNS queries from the last question?

www.google.com

142.250.114.99,142.250.114.103,142.250.114.106,142.250.114.105,142.250.114.104,142.250.114.147

vqaja7xm18.execute-api.us-east-1.amazonaws.com

143.204.130.3,143.204.130.120,143.204.130.86,143.204.130.39

8. Looking at the protocol hierarchy, what is the top UDP protocol in this capture?

DNS over UDP 53

9. Looking at the protocol hierarchy, what is the top TCP protocol in this capture?

TLS over TCP 443

10. Based on the information from the last question, what do you conclude with regards to all TCP communications?

All TCP communications is over TLS which means that all communications is encrypted.

11. Where there any http requests?

Yes, there were a lot of http requests, and they all came from 169.254.144.117 → 239.255.255.250

12. Where there any http requests by a server?

No

13. How many connections were present for ICMP?

0

14. How many connections were present for DNS?

16

15. Did the payload download? If so, look at its contents and tell me what it says.

16. How many connections were present for TLS?

44

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
network-forensics-client-amd64	N/A	Linux executable binary that simulates malware by creating network traffic

This section provides details of the digital evidence collected

Name	network-forensics-client-amd64
Type:	ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=269a44a74666f71f576ef02044ca2f0a1298098a, for GNU/Linux 3.2.0, not stripped
Size:	4036352 bytes
MD5:	5851F2ABE660FFE3CA4A2C23B47EFDDF
SHA1:	6ABD7DEE53E788E2183FC6CF4EFAB0191C0D7092
SHA256:	305B581029B0C405B48483752EE280ED7B6B3B68BC22C829415C0643E6AD96DF
SSDeep:	49152:QyJzRNbtm5fXyyBEikxvAMuU14NZ6V6loa51fA/V8BkKrCuzgnUyC4gZwJrHViJt:Qy9mszq51fAOTvJ4dZ56qqfi+F

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical / Virtual	Built
Greyhatter	Fedora Linux 42 (Workstation Edition)	2025	Virtual	02/06/2026

Software

Name	Version	Release	Purpose
tcpdump	4.99.5	2025	Command-line packet capture tool to record network traffic
tshark	4.6.0	2025	Command-line network protocol analyzer used to read, filter, and analyze packet capture files

Observables

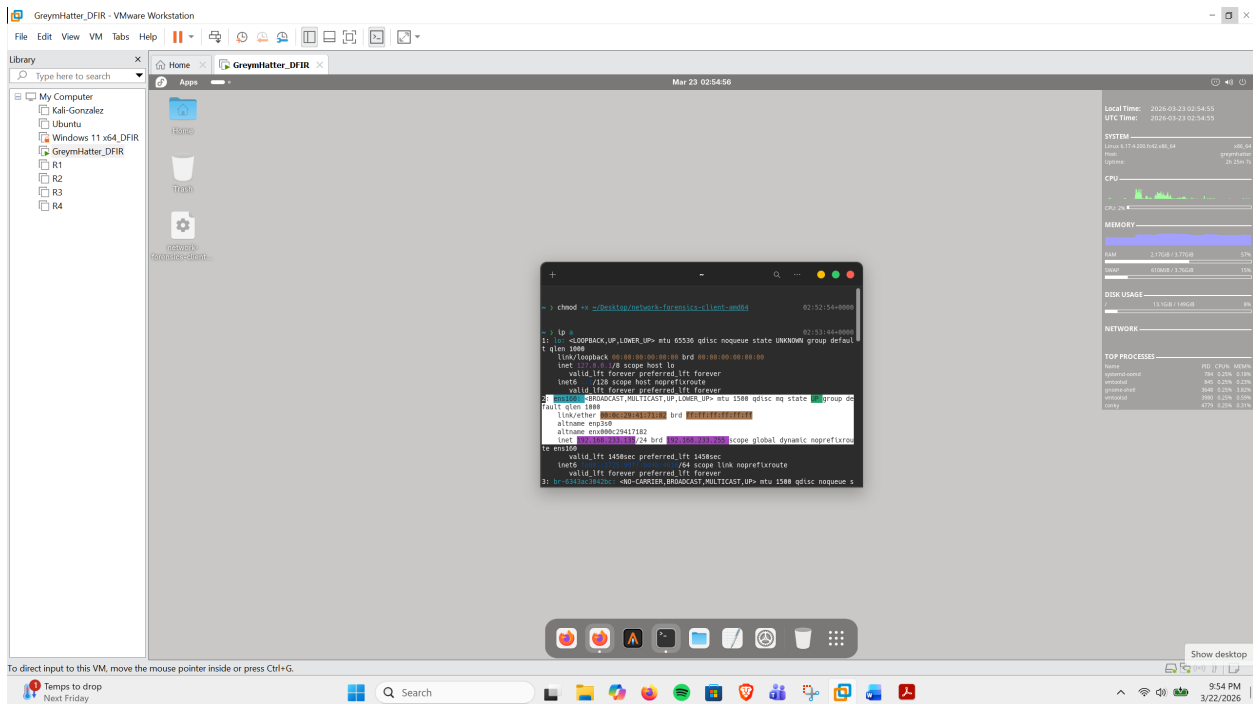


Figure 1: Screenshot of using 'ip a' command to find the interface

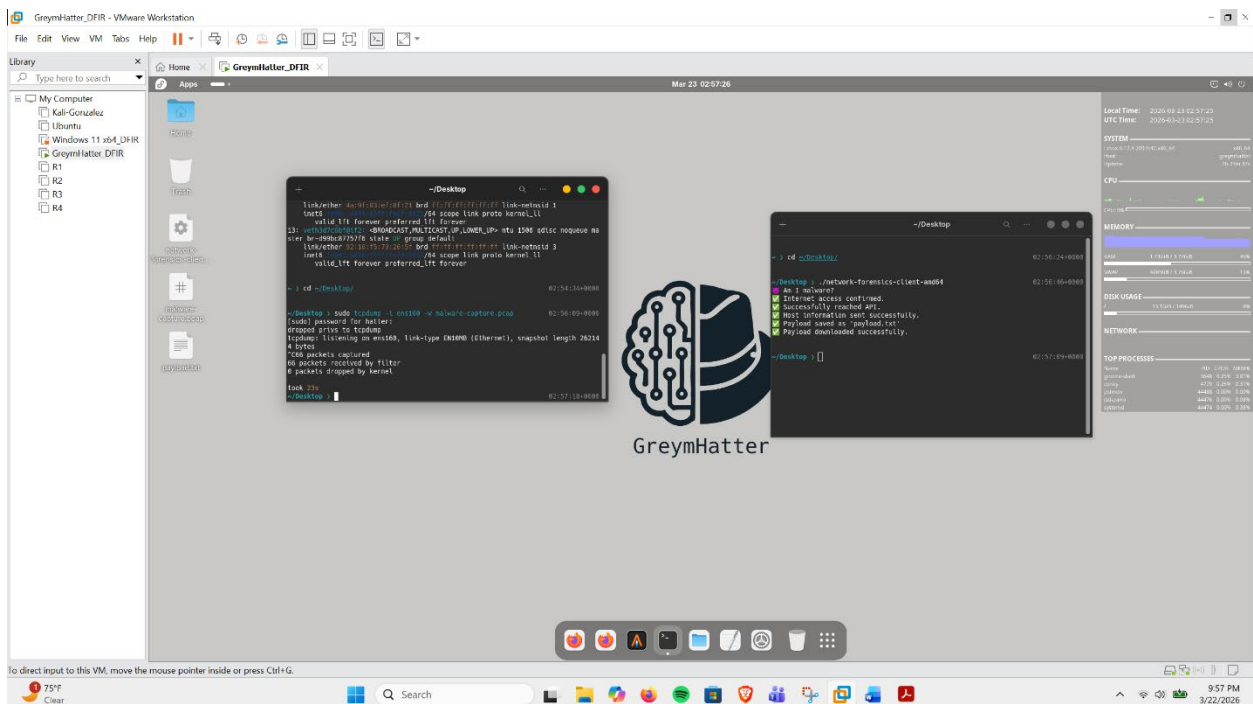


Figure 2: Screenshot of running tcpdump while executing the malicious binary


```

=====
Protocol Hierarchy Statistics
Filter:

frame          frames:66 bytes:29706
  eth          frames:66 bytes:29706
    ip         frames:65 bytes:29500
      udp      frames:21 bytes:2890
        dns    frames:16 bytes:2132
          ssdp frames:4 bytes:572
            mdns frames:1 bytes:186
      tcp      frames:44 bytes:26610
        tls    frames:17 bytes:25034
          tls  frames:1 bytes:1544
    ipv6      frames:1 bytes:206
      udp     frames:1 bytes:206
        mdns  frames:1 bytes:206
=====

```

Figure 6: Screenshot of Protocol Hierarchy from TCP and UDP

```

~/Desktop > tshark -r malware-capture.pcap -Y "http.request"
  3   0.700588 169.254.144.117 → 239.255.255.250 SSDP 143 M-SEARCH * HTTP/1.1
 44   4.745223 169.254.144.117 → 239.255.255.250 SSDP 143 M-SEARCH * HTTP/1.1
 59   7.751989 169.254.144.117 → 239.255.255.250 SSDP 143 M-SEARCH * HTTP/1.1
 62  10.758943 169.254.144.117 → 239.255.255.250 SSDP 143 M-SEARCH * HTTP/1.1

~/Desktop > tshark -r malware-capture.pcap -Y "http.response"

```

Figure 7: Screenshot of http requests

```

~/Desktop > tshark -r malware-capture.pcap -Y "tcp.port==443" | wc -l                                03:39:00+0000
44

~/Desktop > tshark -r malware-capture.pcap -Y dns | wc -l                                          03:42:11+0000
16

~/Desktop > tshark -r malware-capture.pcap -Y "icmp.type==8" | wc -l                               03:42:28+0000
0

```

Figure 8: Screenshot of number of connections made from ICMP, DNS, and TLS

Investigator Procedures

The investigation was conducted within a controlled Fedor Linux 42 virtual machine environment to safely analyze the suspected malicious binary. Prior to execution, all applications and background services were closed to eliminate additional traffic. The binary file 'network-forensics-client-amd64', was executed while network traffic was captured in real time using the *tcpdump* utility. The packet capture saved to the file 'malware-capture.pcap'. The capture was ended after the binary was executed and stopped immediately after.

Following the packet capture, the file was analyzed using *tshark*, the command-line version of Wireshark. Many analysis techniques were used to examine the pcap such as IP and protocol conversations, endpoint statistics, DNS query analysis, and protocol hierarchy. There were a lot of filters applied to inspect traffic, specifically DNS, TCP, UDP, ICMP, and TLS. These procedures allowed for the identification of network communication behavior, protocol usage, and potential indicators of malicious activity from the malicious binary.

Conclusion

Analysis of the captured network traffic revealed that the binary established multiple outbound connections to external servers, even though there were no applications or other communications running during the packet capture. The most significant communication occurring with the IP address '143.204.130.3'. DNS activity was prevalent throughout the capture, with frequent queries to domains such as *via.placeholder.com*, www.google.com, and an Amazon Web Services API endpoint, indicating that the program relied on domain name resolution to facilitate its communications.

The protocol hierarchy analysis showed that DNS was the dominant UDP protocol, while TLS over TCP was the primary TCP protocol observed. This indicates that all TCP communications were encrypted, preventing direct inspection of transmitted data and suggesting an attempt to obscure network activity. Additionally, a temporary high-numbered port 50550 was found in the traffic.

HTTP requests were identified in the capture; however, these originated from a local multicast address (169.254.144.117 → 239.255.255.250) and are consistent with background network discovery protocols rather than malicious activity. No HTTP requests were initiated by a server. No ICMP and no SMB traffic was found. A total of 16 DNS-related connections and 44 TLS-related connections were identified.

Overall, the binary exhibited behavior consistent with simulated malware, including external communication, reliance on DNS resolution, and the use of encrypted channels for data transmission. While the payload was downloaded, analysis of the network traffic did not reveal observable payload content, likely due to encryption. The findings demonstrate typical malware communication patterns while emphasizing the challenges of analyzing encrypted network traffic.

Exhibit A – TCP stream over TLS

```
Node 1: 142.250.114.99:443
517
.....^~.~$.~.;.....q~?.....^.....-.....c..ag.p..uX.....@.....,0.....+./...$.(.k.#.'g.
...9.....3.....=<.5./.....s.....www.google.com.....
.....#.....
.....+.....-.....3.&$.~.:!'.~.r=ZD+?.~&.df.>..D.._HZ.....
.....
2801
...z...v...+...iJ.!S...A.d.R.a...*...^.....^.....-.....c..ag.p..uX.....3.$... V.{.s...Q.7.k..}`)...&Z...'..
(.....
g..D.....{.Z...R[.N..>J$.~.t..u#.~.9.g.....;E;.....Q..o.=.u.[.4cm6"A`.....H.]......|^5.G....}{...n.'..f!..
.\.L....."\.....Ku.{.J..J..0..n'U.....G..A..A.h~.....D...:(.m0
`..CW}.0L.= {c.h.....'u<...H.,EUo...^Y;$.m/.2..`..Z..Y.....S.=.....'r..w.6...P..p.+.....zj.....C.Nd.....'.8
.b.D.....K.....S.....+|w..C..>..4u8.Dlw..C.....+..V.....y...||K.p..n.....+.....0Y?.....$B..!...A.e..t...)>
...0@.4Q.n.....ji..2..Uy..Lst.U...N.....2t..}p.!'"N...b...y.%[.....[P.H'.L.G..7.].0..`.....&..(.;...s...x...W.$T*.
0.o...|0+...?e)..;..a...X..z..f.....).U.rk..nQ.....d...
?..rg..c..t..f.M.0&e..].u.P..^nc.ed...nu..0.....y...p27...R1.....<...jH...F..CE...mx...S...l.q.iB.....$..L.y..a
...f<@..0.....3.r6.....e=s.#.....E.x..p....

,0*.L...w.X...h...^..F.l.H...].M.|.F.J.....?F.BPh2..`-`.....Z.....E.%...rEH..T..Y.....[...{7<...n.A.&|51.T. ....p...de.o
iB...z...t.k.+.....FX..#m..Hc..
...P%...2...>..X..g..v...^..h'.5.A... ..Q /Q.c.w.....Q.....ar.\t..w.aW.Q.../.....L..].R.a"0..K.Q...
>.....!..g.q@...E'.F^...[...=.....m'.~.9f0-T.....2.I^!.!,S.....Z.....K...f...?K...r....
..._OU...Y.....*aP#3S.....B.>.....\e|...Mf.
(-I..m*$...KW...../} .....y.G.>WA.....oo..}.o|.h..E.....+8.....B.B2.0...s..#'.~$.~.unt
.....:0o...s.h.].~.9.e..$.J.....(.A.....zj.....srq.....TB.$p].S<....f<~Q.1. ...<.m
..%..+hbN...6Q.....%V.....nz !Zg$M...@F.....
..{...../?Ln..Q..6.[..=M..p+e...x.gA.n...r..B.
0...>.<.:7.<K...s5...c...6../.F.I....J'.\..y......Pk..d.\G(x."a...g.h.3%..>.Ih.....h~.(A..v...R.+..
:..o.o.3Q.....pI..."/..tR..N.v...VVQ%.q..Q+t.4.0...D;.....I.?x{^.....0'.s}l3(.~/Tn...x..{...UL..7..W)et
h...y...Z..L..m.8...m#..v...0...2A$.!.:x.3q.f...p...>...3.l.`]t'...&..k)*...8qK....|...&..*..i...9.k...N.5.
...kzb^<..J|.Y...K.....dm...#...5h^L.X6R.E..e[.G..~L%...h...n<6,w...+..K..v.0.R.....\..C.Z0z..4)W...].*)<8..(
.....!z.7..79.....l.$..i.;25{o.0b9.t...R.kT.sb..DK.4...v.....Vj..F..u.n.8...`1_E...e..H.y...oc...aI..
.....Q..2.....w&].s...r$.5g-u7|.f.....M7.*..F4.}0..`Y%...X"x..t..0*y.....+..t..JSQ.....K.>=,.....K..p=B|
K.....b..a..J|.e.....P..o..W...N..L((..(-'1...42Y.....UJ.b.....j...E..N.....Hs*@..c.....|&%..-Z.....fS.....;...L
.FK.....
.....j.....&*;d.#(.....hN.../H.....i..~^G..W!...<|.o..C>
80
.....Ey...b.<$..J..t...]"R.
/l.....(.;...:Z...J.6.
75
....F.}.|2..PK..2.q.c.l.....*l.'..?7J.Z...I.4.U.i...2.y..F..Kp.....T...
12560
...7..mo... a.H{(.~.%.4.)8d...k...Ze..bb.W.k.....%.q.W...:..H&A.....*.....P..&=o.kQ...L.VE.%Y...`W..0^U..#...q...
.&.....M...5...lq.M.#.<..R.'.....7...R.$...q\q5.C1.....2+...o-d3f.]N.....uL.V.9.....TA.....
.R.2...U..b.v&..20...rk?...?.....X...x.\.....b...y...t.....zq'(.q..<.....-D.....E..=7..1P.....-.....i.TK.
..m2..b\..#~".....4.....7.....!..l.Hy...>2..a...@i...@U...i.....l4.....(.....
.1.. 'm..&<m..v.P...<Z..L.B[r..r..b..|...U..D...o%...g.P.....Xp5Q?..].2.....z..1.....3.....^..u.?Y
k.a...6.{.C..zSzS]b.I'.....Z....."$|.._0k'F.q.....^4g...0..).H...D.2.....]na(.....2..u'...{.....Jb_AR...]<...
c.N|./y...%.l.U.j.....RE...3...Y.Z...K.....#tZx...A..'.a>e.?..M*.....J..G...o..v..P...z.mqN...'.e..N..b
.%.....h...R.?'..=...V.J.J{g..V5`>.Z9...nW<...v.....!2Gd...3...j(!..l...../..}.g...../.....u.p.....~..~
j{...&A.j.....{R..5.....ce.XR..I..~=.f[M0-...g..H..B.....g...L..(..qQ3[Ph.
k.T..Q_r..z...).P0..&..hoU..`ih..XQ(>u...))...qX..hZ|...{Hv..%9.y...Q.w$.~C.!..ck.3..F...[..jm5.n0D...V.wU..U...
...h.b...Zv!..0...I..^F.$..r...:
.G...M.x.V<!.....Zz1.U...r..p*.S.....P...Mv.....\.....( ...N6+...[...o[...D.m.]eD.X...f..e1.W...=X.N...
[.J.....|.I.I...|.I.....v6.^..y.&x.8...c.#...t...0...MD...9...JH.....^N.pY..*.V%I1..+a...VI..{...>..FA.....8..+
...o..d>...0r...!.(Ok.....t...0.c.[H.....m.Or..F..6Q7+pr..o.Z
..t...<..P...vx..SM.CX...>0..4..W.1..I...g..BG.x!...EYn)s.....>h...5...
...sZ*...am...?IM...?..n...hh\C.W..he.....i.d...r..K..L.<.....=.....*4E*
..5...J)V...w$.C..d.../Z...W...-...6'...oW.Oj@.4...R.Kx..G...y...x...5{.I^./..R.r2.t....&..F.P.4..G.o.YV.....
...V...Di...z..7.o..w.n...).y..W.M.)8.M..xs..'.y./h.B..`64X..N..}.Ws..B..X..t.[...7...~o;...K.+0+...K.....L...
.....^W".Y..".b...).vT...M.1...j.V.Fl.a.....$.F.....D"...xn.p.[.....!JN..k..3H.p..}.x.*".m.+Fzc].$.fB
|S1..Q.....@|.X+o.w...t.z.v...@...V..2.e...Ph.P..G.>vz(u...S_...n...4...R..3.(.t'..gY.e.....0c.{V.U.G.k.0\{3..
h..M...l..J}<...T..Gf4(e..xe.0...h..l..m..W..7..l..#S1".....t...c..e0...&1..a...J.R.s..H...=...t@v...i..h
```

Figure 9: Screenshot of output of TCP stream over TLS of the payload data; payload is not readable and is clearly encrypted